AI-era matrix-multiplication records, decided from their bytes: AlphaEvolve, AlphaTensor, the EinsteinArena table, the 3×3 addition count and the rank walls

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Abstract

A fast matrix-multiplication algorithm is a finite exact object: a rank-R decomposition of the matrix-multiplication tensor, three coefficient matrices whose defining identity, the Brent equations, either holds over a stated ring or does not. We re-decide, from bytes pinned by digest and sharing no code with any producer, the records that AI systems have set or claimed around this object and its neighbours. AlphaEvolve's rank-48 $\langle 4,4,4\rangle$ decomposition is certified over $\mathbb{Z}[i]$: all 4,096 equations hold exactly once the half-Gaussian denominators are cleared. AlphaTensor's rank-47 $\langle 4,4,4\rangle$ is certified over $\mathbb{F}_2$, and the same factors fail over $\mathbb{Q}$ at 465 of the 4,096 equations: the ring is part of the claim. Both walls of the rank of $\langle 3,3,3\rangle$ are re-decided: a rank-23 witness as a certificate, and, of the 496 nodes of Wang's 2026 certificate for the lower bound 20 over $\mathbb{F}_2$, 18 are attacked by an instrument that computes the true rank of each sub-tensor rather than re-running the author's inference rules, 4 proved two-sided, 0 refuted, 478 beyond the instrument. The July 2026 claim of 55 additions around a rank-23 scheme is re-derived gate by gate $(13 + 14 + 28)$ together with its 729 Brent equations. A free flip-graph walk over $\mathbb{F}_2$ reproduces 3 published polynomial-multiplication bounds, one of them tight, and beats none. And the EinsteinArena "new state of the art" table, 20 constructions on 8 problems, is read as the exact rationals its decimals denote: 16 exact witnesses, 4 witnesses only within a verifier's tolerance (the platform's 10^{-9} on circle packings, in force when the packing was published and removed since; the repository notebook's 10^{-6} on three sums) and repaired from their own bytes, no failure, all 8 improvements positive, one printed digit that belongs to a tolerance, and a grid maximum replaced by a certified supremum. No bound is improved. The contribution is the decision grammar — verdict, witness, refusal as a result — and a public record that re-runs in seconds.

1 Introduction

Strassen multiplied two 2×2 matrices in seven multiplications [1]; Winograd proved seven optimal [2]; Laderman multiplied 3×3 matrices in 23 [3], and no one has done better since. Brent wrote the problem as a system of polynomial equations in the coefficients of a bilinear

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algorithm [4], which is the form in which every claim below is decided. The lower bound for 3×3 stood at 19 from Bläser's 2003 argument [5] until March 2026, when Wang's automated method raised it to 20 over $\mathbb{F}_2$ [6]. Two AI systems entered the upper-bound side: AlphaTensor [7] produced, among other factorisations, a rank-47 algorithm for 4×4 matrices in modular arithmetic, and AlphaEvolve [8] a rank-48 algorithm over the complex numbers, presented as an improvement on the recursive Strassen rank 49. Flip-graph search [10] is now the standard non-learned competitor, and has been pointed at polynomial multiplication [11]. The additive cost of Laderman-rank schemes became an active record in 2025–26, ending (for now) at 55 additions [12]. And Together AI's EinsteinArena "new state of the art" table claimed constructions on classical extremal problems that beat AlphaEvolve's, each checked in floating point, with tolerances, by the notebook in its repository [13]; the EinsteinArena platform's own verifiers score in floating point too, under rules that have moved since [14].

Every one of these results was verified by its producer. The table's notebook checks in floating point with stated tolerances; the DeepMind notebook verifies its own decomposition in a cell; the lower-bound certificate was checked by the code that produced it. None of that is a defect, but it leaves a specific gap: no independent party had decided these objects in exact arithmetic, from the bytes as published, with a stated verdict grammar and a record anyone can re-run. This paper fills that gap for the objects named in the title and reports what was found, including the two findings that are not confirmations: a printed digit that belongs to a tolerance rather than to a witness, and a benchmark score that is a grid maximum standing in for a supremum. Nothing here improves a bound; a REPAIRED construction is not a refutation; a lower-bound node not attacked is not a doubt. The novelty claimed is the grammar in which each object is decided and the record underneath it, not the arithmetic, which is elementary throughout.

The kissing-number rows of the same platform's table are decided in a separate paper [23] and are not repeated here.

2 The object, the ring, and the grammar

Let A be $n \times m$ and B be $m \times p$. Index the entries of A by $ab = a \cdot m + b$, those of B by $bc = b \cdot p + c$, and those of $C = AB$ by an index k under one of two layouts, AC with $k = a \cdot p + c$ or CA with $k = c \cdot n + a$. A bilinear algorithm of rank R over a ring $\mathcal{R}$ is a triple of matrices $U \in \mathcal{R}^{nm \times R}$, $V \in \mathcal{R}^{mp \times R}$, $W \in \mathcal{R}^{np \times R}$, run as

$$L_t = \sum_{ab} U_{ab,t} A_{ab}, \qquad R_t = \sum_{bc} V_{bc,t} B_{bc}, \qquad C_k = \sum_{t=1}^{R} W_{k,t}\, L_t R_t.$$

It is correct for every pair of matrices over $\mathcal{R}$ exactly when the one finite identity

$$\sum_{t=1}^{R} U_{ab,t}\, V_{b'c,t}\, W_{k,t} \;=\; [\, b = b' \,] \cdot [\, k = \mathrm{idx}(a, c) \,] \qquad \text{for all } ab,\ b'c,\ k \tag{1}$$

holds: $nm \cdot mp \cdot np$ equations, each an exact sum of R products (the Brent equations [4]). The layout is a publishing convention, not mathematics; the instrument tries both and records the one under which (1) holds, and fails a claim only when it holds under neither.

Exactness. Every published coefficient here is a small integer, so each sum in (1) is bounded by $R \cdot \max|U| \cdot \max|V| \cdot \max|W|$; the instrument checks that this bound lies below 2^{53} before summing in doubles, which are then exact, and its battery cross-checks a full audit against an arbitrary-precision evaluation. Anything past the bound is refused, never approximated. Over $\mathbb{F}_2$ a sum is a parity. Over $\mathbb{Z}[i]$ a coefficient is a pair of integers and a claim may carry an integer scale s: the identity decided is $\sum_t U_{ab,t}V_{b'c,t}W_{k,t} = s \cdot [b = b'][k = \text{idx}]$ with imaginary part exactly zero. This is the natural home of a decomposition published over $\frac{1}{2}\mathbb{Z}[i]$: doubling the factors clears the denominators and the claim becomes $\sum(2u)(2v)(2w) = 8T$.

The word *certified*. We say that a claim is *certified* when its defining finite identity has been decided in exact arithmetic over the stated ring, from bytes pinned by their SHA-256 digest, by an instrument of this repository whose red controls (planted forgeries that must be refuted) fire in the same run. Everywhere else we write *decided* for the act and *verified* or *refuted* for the outcome; "proved" is reserved for statements with a proof in the usual sense. The grammar of a verdict is fixed per instrument and stated with it below: for a decomposition, VERIFIED, REFUTED (with the first failing equation named), REJECT (the identity holds but the rank is not below the naive *nmp*), and REFUSED (not decidable as given: a failed pin, a bound past exactness). A refusal is a result and is reported as one.

Statement of results. The following are decided on the machine that built this document, at repository commit `9349d8bad505`, from the records named in the reproduction section.

Proposition 1 (AlphaEvolve)**.** *The rank-48 $\langle 4,4,4\rangle$ decomposition in the DeepMind notebook pinned at commit `4226acb` (digest `2cce2543e48c89aa…`) satisfies* (1) *over $\mathbb{Z}[i]$ with scale* 8 *under layout CA: all 4,096 equations hold exactly, every imaginary part vanishing. It is certified; its rank is below the generated Strassen $\otimes$ Strassen rank 49, itself re-decided from scratch.*

Proposition 2 (AlphaTensor, both rings)**.** *The rank-47 $\langle 4,4,4\rangle$ factors of the pinned `alphatensor_f2.npz` (digest `70f09f349d8d2874…`) satisfy* (1) *over $\mathbb{F}_2$ (all 4,096 equations, layout CA) and are certified there. The same factors, read as integers, fail* (1) *over $\mathbb{Q}$ at 465 of the 4,096 equations (11.4%), the first at $(ab, b'c, k) = (0, 0, 2)$ with sum 2 where 0 is required. The rank-96 $\langle 5,5,5\rangle$ factors of the same file behave the same way: certified over $\mathbb{F}_2$, failing over $\mathbb{Q}$ at 1,043 of 15,625 equations.*

Proposition 3 (the 3×3 walls)**.** *A rank-23 $\langle 3,3,3\rangle$ decomposition (AlphaTensor's, from the pinned `alphatensor_r.npz`, digest `4d59571a2537a947…`) is certified over $\mathbb{Q}$: all 729 equations hold. Of the 496 nodes of Wang's certificate for $R_{\mathbb{F}_2}(\langle 3,3,3\rangle) \geq 20$ (digest `25595a883ce877ee…`, which equals the git-lfs object id the author published), 18 were attacked by computing the true minimum rank of the node's constrained sub-tensor over $\mathbb{F}_2$: 4 are proved two-sided, 8 carry an upper bound equal to the claimed lower bound, 6 survive a search for a cheaper decomposition, and 0 are refuted. 478 nodes (96.4%) are beyond the instrument and rest on the author's reasoning.*

Proposition 4 (the addition count)**.** *The certificate of Karunaratne and Idamekorala [12] (digest `91c36b1165a71d4b…`) is verified: its three straight-line programs compute exactly the factor matrices they claim, with* 13, 14 *and* 28 *gates,* 55 *in all, as declared; every coefficient is in $\{-1, 0, 1\}$; and the factor matrices satisfy all 729 Brent equations over $\mathbb{Q}$ (layout AC) at rank 23.*

Proposition 5 (polynomial multiplication over $\mathbb{F}_2$)**.** *9 bilinear schemes found by a free flip-graph walk are certified over $\mathbb{F}_2$ against targets rebuilt by literal polynomial arithmetic. Against the published table, 3 match the best known upper bound (`C7`, `T6`, `T7`; `C7` is tight), 3 land above it, 0 beat it, and 3 match the Chen–Kauers full-product table.*

Proposition 6 (the EinsteinArena table)**.** *Of the 20 constructions the repository publishes at commit `c388c6f7`, every one read as the exact rationals its decimal literals denote, 16 are exact witnesses of the value stated, 4 are witnesses only within a verifier's tolerance (the platform's, as it stood when the construction was published, or the repository notebook's) and are repaired from their own bytes, and 0 fail. All 8 improvements over the previous best are positive exact differences, from 9.06×10^{-9} to 5.99×10^{-2}. Of 19 printed values, 18 are the exact value's rounding or ceiling; the one exception is the tolerance's digit.*

3 The registry: AlphaTensor and AlphaEvolve

Table 1 is the corpus of `families/strassen-audit`, re-certified in-process while this document's macros were written (32 ms for all 11 rows, every source pin re-hashed): 10 verified rows holding 46,466 equations between them, and 1 honest reject. Strassen's 1969 scheme, transcribed from the textbook, is the calibration; Strassen $\otimes$ Strassen is generated by exact Kronecker composition and re-decided as a fresh claim; the naive rank-8 algorithm, the definition itself, certifies as correct and is REJECTED because a hit asserts a rank strictly below nmp. Correct is not fast. The 5 AlphaTensor rows over $\mathbb{Q}$ and 2 over $\mathbb{F}_2$ are converted mechanically from the two pinned `npz` files of the DeepMind repository; the C-index layout differs between Strassen's transcription (AC) and the npz factors (CA), and the audit detects it rather than assuming it.

row	shape	rank	naive	ring	equations	layout	verdict	source
`strassen-1969`	$2\times2\times2$	7	8	$\mathbb{Q}$	64	AC	VERIFIED	Strassen 1969, textbook transcription
`strassen-squared-4x4x4`	$4\times4\times4$	49	64	$\mathbb{Q}$	4,096	AC	VERIFIED	Strassen $\otimes$ Strassen, generated here
`naive-2x2x2`	$2\times2\times2$	8	8	$\mathbb{Q}$	64	AC	REJECT (correct, not fast)	the definition, generated here
`alphatensor-q-2x2x2`	$2\times2\times2$	7	8	$\mathbb{Q}$	64	CA	VERIFIED	AlphaTensor npz, key 2,2,2
`alphatensor-q-3x3x3`	$3\times3\times3$	23	27	$\mathbb{Q}$	729	CA	VERIFIED	AlphaTensor npz, key 3,3,3
`alphatensor-q-4x4x4`	$4\times4\times4$	49	64	$\mathbb{Q}$	4,096	CA	VERIFIED	AlphaTensor npz, key 4,4,4
`alphatensor-q-3x4x5`	$3\times4\times5$	47	60	$\mathbb{Q}$	3,600	CA	VERIFIED	AlphaTensor npz, key 3,4,5
`alphatensor-q-4x5x5`	$4\times5\times5$	76	100	$\mathbb{Q}$	10,000	CA	VERIFIED	AlphaTensor npz, key 4,5,5
`alphatensor-f2-4x4x4`	$4\times4\times4$	47	64	$\mathbb{F}_2$	4,096	CA	VERIFIED; refuted over $\mathbb{Q}$	AlphaTensor npz, key 4,4,4
`alphatensor-f2-5x5x5`	$5\times5\times5$	96	125	$\mathbb{F}_2$	15,625	CA	VERIFIED; refuted over $\mathbb{Q}$	AlphaTensor npz, key 5,5,5
`alphaevolve-48-4x4x4`	$4\times4\times4$	48	64	$\mathbb{Z}[i]$	4,096	CA	VERIFIED	AlphaEvolve notebook @ 4226acb

Table 1: Every row of the fast-matmul corpus, re-decided from pinned bytes. "Naive" is nmp, the schoolbook count; "equations" is $nm \cdot mp \cdot np$, every one decided exactly.

AlphaEvolve's 48, from mutable bytes. The only first-party byte source for AlphaEvolve's decomposition is a Jupyter notebook in a DeepMind repository whose main branch has been rewritten since the result was announced; the arXiv paper [8] carries no ancillary files and is the citation, not the source. The audit therefore pins by commit: the notebook is held at digest `2cce2543e48c89aa…`, re-hashed at every certify, and the row refuses if a byte moves. Every published entry lies in $\frac{1}{2}\mathbb{Z}[i]$ — the doubled factors use 9 distinct coefficients from $\{0, \pm1, \pm i, \pm1 \pm i\}$, 1,376 of the 2,304 entries non-zero — and $\pm\frac{1}{2}$ is exact in the notebook's `float32`, so the byte-level parse loses nothing. The certified claim is the scale-8 identity of Proposition 1. Its converter audits the claim before writing the corpus row, so a mis-parse cannot land in the record; the row is then decided again, from the record, by the family.

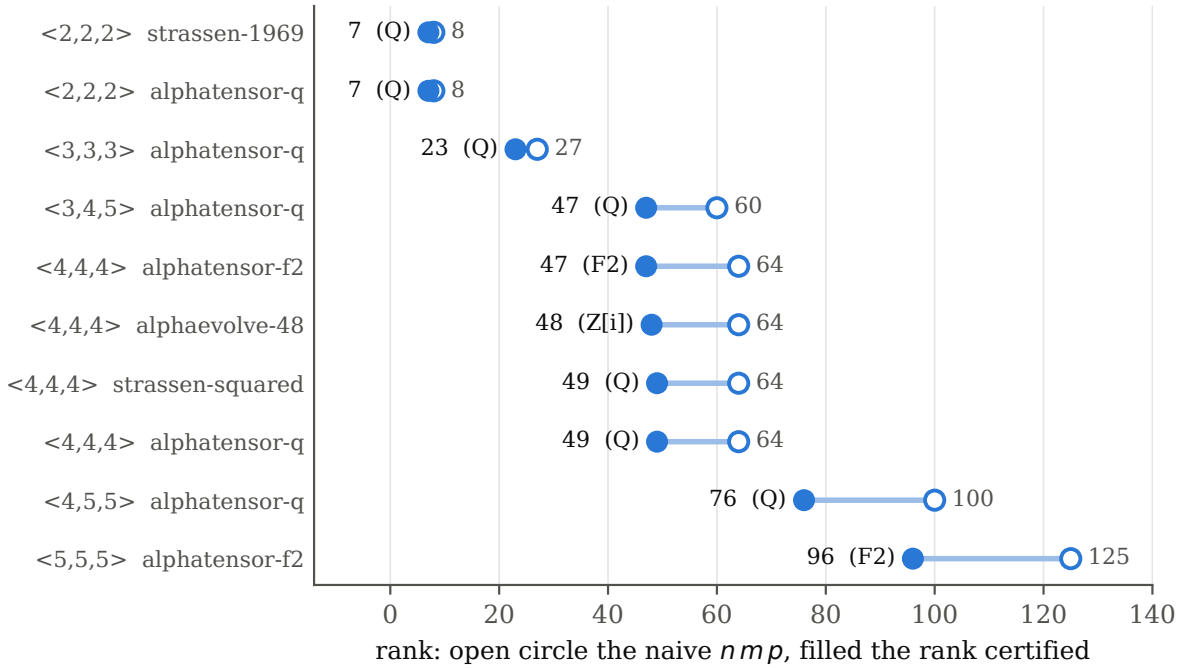


Figure 1: The ten verified rows, each drawn from the naive rank of its shape to the rank certified, with the ring beside it. The ring is not decoration: the two $\mathbb{F}_2$ rows are refuted over $\mathbb{Q}$.

AlphaTensor's 47, decided both ways. The Nature paper says that its rank-47 4×4 algorithm works in modular arithmetic [7]. The audit turns that statement into a decided pair: the same pinned factors are verified over $\mathbb{F}_2$ and refuted over $\mathbb{Q}$, with the failing equations located (Proposition 2). Every one of the 4,096 integer sums agrees with its target modulo 2, as it must, and 465 of them differ as integers, by as much as 8. The speedup over Strassen $\otimes$ Strassen's 49 therefore does not survive lifting to characteristic zero — a statement about the published artifact, produced mechanically, of a kind a benchmark run cannot make, since running the algorithm on integer matrices would return answers that are wrong and correct modulo 2. The AlphaTensor $\langle 4, 4, 4 \rangle$ factors over $\mathbb{Q}$ in the same release have rank 49, the Strassen-squared rank, and are verified as such. We do not call the $\mathbb{F}_2$ refutation a correction of anything: the paper states the ring; the record now decides it.

Independence. The whole corpus detaches to `certs/strassen-certificate.json`, and `tools/verify_strassen.py`, written against the standard library only and sharing no line with the instrument, re-decides every entry from the file alone: 10 entries, 0 failures, and a red control that forges one coefficient by +1 and must be refuted before the script exits green. The instrument's own battery runs 34 checks with 6 red controls.

4 The rank walls of $\langle 3, 3, 3 \rangle$

The rank of 3×3 matrix multiplication is the oldest open interval in the subject: the upper wall is Laderman's 23 [3], the lower wall was Bläser's 19 for twenty-three years [5]. Wang's

preprint [6] raises the lower wall to 20 over $\mathbb{F}_2$ by an automated case split whose proof is a machine-checkable certificate: 496 nodes, each a constrained sub-tensor with a claimed lower bound derived by one of four inference rules. So far as we could establish, that certificate had not been independently checked. (Bläser's 19 is a statement over any field; the 20 is an $\mathbb{F}_2$ statement and does not supersede it elsewhere. Border rank is a different quantity and is never quoted here as a rank bound.)

The upper wall is a certificate: the rank-23 decomposition of Table 1, 729 equations over $\mathbb{Q}$. We note that the pinned witness is AlphaTensor's 3×3 factorisation rather than a transcription of Laderman's own table; it has Laderman's rank, which is what the wall needs.

The lower wall is where the method matters. A verifier that re-implements the author's four inference rules can only ever agree with him. The instrument `instruments/tensorlb` instead computes ground truth: for each node whose constrained sub-tensor it can reach, the true minimum rank over $\mathbb{F}_2$ (two-sided, by matrix rank, where the node has dimension at most two), or an upper bound found by search, and asks whether the author's number is the truth. Two conventions the certificate never documents — the semantics of its constraint functionals and the transposed C-index — were recovered from the bytes. The verdicts are kept distinct: CONFIRMED (rank equals the claimed bound, proved both ways, the author's rules not relied on), TIGHT-IF (an upper bound equal to the claimed lower bound was found, which is not a proof of the lower bound), SURVIVES (a search for a cheaper decomposition found none), NOT-ATTACKED (beyond the instrument), REFUTED (a decomposition below the claimed bound exists).

certificate	nodes	witnesses	confirmed	tight-if	survives	not attacked	refuted
$\langle 2,2,2\rangle$ control (rank 7 optimal)	10	10 / 10	3	3	1	3	0
$\langle 3,3,3\rangle$, the bound ≥ 20	496	—	4	8	6	478	0

Table 2: Both of Wang's certificates, re-run here (2.9 s together). The control's root node is the unconstrained $\langle 2,2,2\rangle$ tensor with claimed bound 7, optimal since [2]: a rank-6 decomposition found there would have meant a broken search, not new mathematics.

Table 2 is the result. On the $\langle 2,2,2\rangle$ control (digest `27b0dc70d76e6405…`) all 10 upper-bound witnesses re-verify against sub-tensors rebuilt here, 3 bounds are proved two-sided and 0 refuted. On the target, the $\langle 3,3,3\rangle$ certificate carries no upper-bound witnesses at all; it is a pure lower-bound argument, which is exactly why ground truth rather than rule replay is the right audit. 18 nodes were attacked, 4 proved two-sided, 0 refuted; 478 nodes, those of dimension above two, are beyond the instrument and still rest entirely on the author's reasoning. That limit is the honest content of Proposition 3: the first independent check of part of a new lower bound on a fifty-year-old problem, a reproduction and not a find. The battery runs 12 checks with 3 red controls (an inflated bound, an altered witness coefficient, a deleted witness term), all of which fire.

5 Rank is not the cost: the 55 additions

A rank-23 scheme for 3×3 still has to form its 23 left factors from the nine entries of A, its 23 right factors from those of B, and combine 23 products into the nine entries of C. Those

three linear maps are where the additions live, and the additive record around Laderman's rank moved repeatedly in 2025–26, as recorded on the report page with each step's arXiv identifier, to the 55 of [12], a preprint that was a month old when audited. Its certificate is a format of its own: three straight-line programs of two-input add/subtract gates, the factor matrices they realise, and declared counts.

A rank claim and an addition claim fail differently, so both halves are decided separately. `instruments/slp` evaluates each program symbolically over the integers on the standard basis and compares, row by row, the exact coefficient vector every output computes with the factor matrix the certificate says it realises; it counts the gates itself and trusts no declared field — a declared count that disagrees with its own gate list is REFUTED, not corrected. The bilinear half is handed to the tensor instrument of Section 3, which decides the 729 Brent equations. Table 3 is the result: every circuit realises its map exactly, the gates count to 55 here as declared, and the factors multiply 3×3 matrices at rank 23. The total is 78 scalar operations. The same three maps with no sharing at all would cost 122 additions, a number we compute rather than quote; the certificate names its source as a 58-addition scheme (`3x3x3_m23_cr58_cn122_ZT_reduced.json`, side costs $14+15+29$), and derives its output circuit by transposition, $14+23-9=28$. Every coefficient is in $\{-1,0,1\}$, which is what supports the authors' claim that the scheme works over any associative ring; a single 2 would void it, and the battery plants one.

program	what it computes	gates declared	gates counted	no sharing	realises its matrix
`U_input`	form the 23 left factors from the 9 entries of A	13	13	43	exactly
`V_input`	form the 23 right factors from the 9 entries of B	14	14	30	exactly
`W_output`	combine the 23 products into the 9 entries of C	28	28	49	exactly
total	the whole linear cost	55	55	122	agree

Table 3: The three linear maps of the 55-addition certificate, each decided on its own. "No sharing" is the cost of the same map with no common subexpressions.

Two things are not decided here and the authors are careful about both. Their optimality statement is local: the certificate records exhaustive searches giving lower bounds of 13 and 14 gates for the two input maps in their fixed orientation of this fixed tensor, which this audit reads but does not re-run; and nothing is claimed about the minimum over all rank-23 schemes, which is open. The battery: 16 checks, 8 red controls (a flipped sign, a deleted gate, a forged per-circuit count, a forged total, a gate reading a slot not yet written, a gate out of order, a perturbed factor coefficient, a coefficient of 2), all firing.

6 Polynomial multiplication over $\mathbb{F}_2$: a free walk against the published table

The same certificate shape covers any bilinear map. For the product of two polynomials with n coefficients over $\mathbb{F}_2$ — full (P_n, $2n-1$ coefficients out), truncated (T_n, the low n) and cyclic (C_n, modulo X^n-1; over $\mathbb{F}_2$ the negacyclic product is the same tensor) — the best upper bounds are hand constructions from 1977 to 2011 [17, 18, 19, 20, 21, 22], and every lower bound beneath them moved in Wang's 2026 preprint [6]. Flip-graph search had been run on the full product by Chen and Kauers [11], whose square cases land exactly on the published bounds; and Wang himself, in a footnote to his table, improved T_9 from 27 to 26 by flip-graph search over $\mathbb{F}_2$ — the one truncated case modern search had touched. That prior art was read before the search was

written, and it turned the task from an attack on the full product into a sweep of the families nobody had swept, with the Chen–Kauers table as a calibration ladder.

The proposer is a random walk on the flip graph with the rank-increasing "plus" transition, free of any model or API call; a flip preserves the tensor identity, so the walk screens by rank and never by correctness, and the instrument `instruments/bilinear` decides every scheme the walk reports. The instrument rebuilds the target from its *name*, by literal convolution and reduction, and never accepts the claimant's tensor alongside the claim: any scheme decomposes some tensor, and a certifier that takes the target on trust checks a tautology. Its battery puts Strassen's rank 7 through this instrument and the older one and requires agreement, and refutes a correct C_7 scheme audited as T_7.

target	family	naive	published (lower–upper)	walk	standing	upper bound due to
`C7`	cyclic, $n = 7$	49	13–13	13	MATCH	Wagh–Morgera 1983
`C8`	cyclic, $n = 8$	64	19–22	25	above	Morgera 1990; Cenk–Özbudak 2011
`P2`	full, $n = 2$	4	CK 3	3	matches CK	Chen–Kauers 2025 table
`P3`	full, $n = 3$	9	CK 6	6	matches CK	Chen–Kauers 2025 table
`P4`	full, $n = 4$	16	CK 9	9	matches CK	Chen–Kauers 2025 table
`T6`	truncated, $n = 6$	21	13–14	14	MATCH	Cenk–Özbudak 2011
`T7`	truncated, $n = 7$	28	16–18	18	MATCH	Cenk–Özbudak 2011
`T8`	truncated, $n = 8$	36	19–22	23	above	Cenk–Özbudak 2011
`T9`	truncated, $n = 9$	45	21–26	27	above	Wang 2026, footnote

Table 4: Every target certified, against the literature as transcribed in the bounds corpus (retrieved 2026-08-30). "CK" is the Chen–Kauers full-product rank. Each walk ran 8 or 12 seeds for 300,000 or 800,000 steps at zero cost.

Table 4 is the result, and it is a null one, reported as such. The walk reproduced 3 published bounds from the naive algorithm, told nothing (`C7`, `T6`, `T7`); at `C7` the walls meet, so the rank 13 it reaches from the naive 49 in 20 s is the exact rank. It landed above the literature at `C8` (25 against 22), `T8` (23 against 22), `T9` (27 against 26), and beat nothing. The 9 stored schemes are re-decided at every build (2,824 equations over $\mathbb{F}_2$ this run); the whole campaign cost 1,331 s of one laptop core. 6 published targets were never attempted (Table 5): the campaign was stopped, not exhausted. One refusal is wired in and has never fired: a scheme certifying below a published lower bound refuses the build outright, because it would mean a 2026 theorem is false and it is far likelier that this repository is wrong. The battery runs 16 checks, 8 red controls and a 4-rung ladder of published ranks the walk must still reach at a fixed seed.

target	lower (Wang 2026)	previous lower	upper	upper bound due to
`P6`	16	15	17	Montgomery 2005
`P7`	19	18	22	Montgomery 2005
`P8`	21	20	26	Winograd 1977; Fan–Hasan 2007
`C10`	22	19	29	Morgera 1990
`T10`	24	20	31	Cenk–Özbudak 2011
`T11`	27	22	36	Cenk–Özbudak 2011

Table 5: The published targets this campaign did not attempt.

7 The EinsteinArena table, re-decided

The repository `github.com/togethercomputer/EinsteinArena-new-SOTA` claims a new state of the art over AlphaEvolve on a table of extremal problems: circle and hexagon packings, Heilbronn configurations, step functions for two open inequalities, a flat polynomial, a benchmark score [13]. Every construction was checked in floating point, with tolerances, by the notebook the repository ships; the platform's own verifiers, read here from their sources at `9cd6fbfb` and dated by their history [14], score in floating point under rules that have moved since the table was published. The ledger of `instruments/easota` reads each published file (36 files held byte for byte at commit `c388c6f7`, fetched 2026-09-08, re-hashed at every run) as the exact rationals its decimal literals denote — 0.1 is $\frac{1}{10}$, never the nearest double — and decides the platform's objective with no tolerance at all. The grammar per row: WITNESSED (the bytes are an exact witness of the value stated); REPAIRED (a witness only within a verifier's tolerance (named and dated in platformRule); an exact witness built from the same bytes, its deficit printed); UNWITNESSED (the bytes fail exactly and no repair was built — a finding about the bytes, never a refutation of the bound). Improvements over the previous best are decided as exact signs between two published constructions. Where a published construction is not decidable as stated, the row says so and why.

problem	construction	printed	exact, 16 digits	verdict	the printed digits
circles in a rectangle	AlphaEvolve V2, 2025-11	2.3658321334	2.3658321334167631	WITNESSED	the exact value's rounding
circles in a rectangle	Together AI agents, 2026-04	2.3658323759	2.3658323758318174	REPAIRED	NOT the witness's rounding
Heilbronn, convex region	AlphaEvolve V2, 2025-11	0.0278355715	0.0278355714584821	WITNESSED	the exact value's rounding
Heilbronn, convex region	Together AI agents, 2026-04	0.0278355805	0.0278355805175536	WITNESSED	the exact value's rounding
min-distance ratio, 2-D	AlphaEvolve, 2025-06	12.889266	12.8892661120346322	WITNESSED	the exact value's rounding
min-distance ratio, 2-D	Together AI agents, 2026-03	12.889230	12.8892299077175173	WITNESSED	the exact value's rounding
Erdős minimum overlap	Haugland, 2016	0.380927	0.3809268534330869	WITNESSED	the exact value's ceiling
Erdős minimum overlap	AlphaEvolve, 2025-06	0.380924	0.3809230351084501	REPAIRED	the exact value's ceiling
Erdős minimum overlap	TTT-Discover, 2026-01	0.380876	0.3808753232177187	REPAIRED	the exact value's ceiling
Erdős minimum overlap	Together AI agents, 2026-03	0.380871	0.3808703105862199	REPAIRED	the exact value's ceiling
edges vs triangles	AlphaEvolve V2, 2025-11	−0.712494	−0.7124938782214395	WITNESSED	the exact value's rounding
edges vs triangles	Together AI agents, 2026-03	−0.712256	−0.7122563326979318	WITNESSED	the exact value's rounding
first autocorrelation	AlphaEvolve, 2025-06	1.50529397	1.5052939684401608	WITNESSED	the exact value's ceiling
first autocorrelation	AlphaEvolve V2, 2025-11	—	1.5031635546815609	WITNESSED	none printed
first autocorrelation	TTT-Discover, 2026-01	1.50286290	1.5028628982558265	WITNESSED	the exact value's ceiling
first autocorrelation	Together AI agents, 2026-03	1.50286286	1.5028628587049711	WITNESSED	the exact value's ceiling
flat polynomial	AlphaEvolve V2, 2025-11	1.340925	1.3409252804007615	WITNESSED	the supremum's rounding
flat polynomial	Together AI agents, 2026-03	1.280932	1.2809320528750416	WITNESSED	the supremum's rounding
hexagons in a hexagon	AlphaEvolve V2, 2025-11	3.9419123	3.9419123000000000	WITNESSED	exact decimal; packing certified
hexagons in a hexagon	Together AI agents, 2026-04	3.9416523	3.9416523000000000	WITNESSED	exact decimal; packing certified

Table 6: The 20 constructions, decided. "Printed" is the number the repository's README states; "exact" is the platform's objective on the bytes with no tolerance (for the two flat polynomials, the lower end of a certified enclosure of the supremum). A score is confirmed when it is the exact value's rounding; an upper bound when it is the exact value's ceiling.

The repairs, and the digit that belongs to the tolerance. When Together published its packing (2026-04-02), the platform accepted a circle packing if no two circles overlapped by more than 10^{-9} and the bounding box's width plus height exceeded 2 by no more than 10^{-9} — the rule the repository's notebook still applies. On 2026-08-24 the platform removed that slack (its changelog, "Feasibility checks tightened"; the code merged 2026-08-31): the box is now checked in exact rationals and the pairs with no margin, and its live verifier, read on 2026-10-06, rejects this file [14]. Read exactly, the Together 21-circle packing overlaps in 44 of its 210 pairs, the worst (pair 3–17) by 3.09×10^{-12} in squared distance, and its box exceeds the perimeter by 4.20×10^{-14}.

AlphaEvolve's packing does neither (worst pair clear by 5.04×10^{-9}, box by 3.20×10^{-8}) and is an exact witness of 2.3658321334. The repair is the smallest the bytes allow: every radius scaled by one rational $\lambda = 0.999999999963354125\ldots$, the largest with $\lambda^2(r_i+r_j)^2 \le d_{ij}^2$ on every pair and the box inside the perimeter. The exact witness then sums to 2.3658323758318174, a deficit of 8.67×10^{-11} against the published sum 2.3658323759185154. The printed 2.3658323759 rounds the published sum, not the witness, which rounds to 2.3658323758: ten digits were printed and the tenth is the tolerance's. The improvement over AlphaEvolve is unaffected, 2.42×10^{-7} between the two exact witnesses. The other three repairs are float noise made visible: the repository's notebook requires a step function's values to sum to $n/2$ within 10^{-6}, while the platform's verifier has no tolerance on the sum — in every public version (since 2026-03-26) it rescales the values to $\sum h = n/2$ in floating point before scoring [14]. Three of the four minimum-overlap constructions miss $n/2$ by 9.9×10^{-16} to 1.5×10^{-14} (Haugland's 51-step function [16], written as short decimals, sums exactly) — inside the notebook's 10^{-6}, and below what a float64 sum can see, so the platform's rescale leaves all three as published — and renormalised to $\sum h = n/2$ exactly the bounds move by at most 8.4×10^{-18}; every printed ceiling stands, 0.380876 and 0.380871 being the ceilings of 0.3808753232 and 0.3808703105.

A supremum, not a sample. The two inequalities are the rows where exactness is more than bookkeeping. A step function's autoconvolution is piecewise linear with breakpoints at multiples of the step, so the platform's discrete maximum is the true supremum of $f * f$ and the printed number is the bound itself. Together's 30,000-value function attains its maximum on a plateau: 10 indices lie within the float screen's stated forward-error bound (2.7×10^{-11}, relative) of the float maximum, every one was decided in exact integers, and the maximum sits at index 28,879 with $C_1 = 1.5028628587049711$, whose ceiling to eight digits is the printed 1.50286286. TTT-Discover's function gives 1.502862898255…, ceiling 1.50286290; the difference, 3.96×10^{-8}, is real. One file in that folder carries no printed number: AlphaEvolve V2's 1,319-value function decides to 1.5031635546…. The flat polynomial is the row where the platform's verifier is not the mathematics: it scores a ±1 polynomial by the largest $|g|$ over a million equally spaced points of the unit circle, a lower bound on the supremum however fine the grid. Here $|g(z)|^2$ on the circle is written as the integer cosine polynomial its autocorrelations define, reduced to degree 69 in $\cos\theta$, and its maximum on $[-1,1]$ is certified by the Sturm-chain and interval-Newton instrument that decides the Chowla cosine minima elsewhere in the repository: 43 critical points isolated and enclosed, the value at each bounded exactly, in 949 ms. The supremum is $C^+ \in [1.280932052875041675,\ 1.280932052875041754]$ (width 7.9×10^{-17}); the grid score the repository prints to sixteen digits, 1.2809320527987995, falls short of it by 7.62×10^{-11}, and the six printed digits, 1.280932, are the supremum's. AlphaEvolve's polynomial certifies to $[1.3409252804007615,\ 1.3409252804007617]$ (its grid short by 9.45×10^{-10}); the two enclosures are disjoint, so the improvement is decided at 5.99×10^{-2} or more.

Hexagons, in intervals. The hexagon packings are the rows exact rationals cannot reach: each of twelve unit hexagons is rotated by a published angle in degrees, so every vertex is a cosine and a sine of a decimal. They are decided in outward-rounded interval arithmetic with certified π, sine and cosine — every literal entering as the two doubles around the exact rational it denotes — so a SEPARATED pair or an INSIDE vertex holds for the exact real configuration. Together's packing: all 66 pairs certified separated by the separating-axis theorem, the closest by at least 3.6×10^{-8}; all 72 vertices certified inside, the tightest with a cross product of at least

4.0×10^{-7}. The platform's rule (a pair counts as intersecting unless separated by more than 1e-9; a vertex counts as inside unless outside by more than 1e-9) was not needed. AlphaEvolve's packing certifies the same way (closest pair at least 6.9×10^{-6} apart); both outer sides, 3.9416523 and 3.9419123, are exact decimals, so the improvement 2.6×10^{-4} is exact. The enclosures are 6.3×10^{-15} wide; a pair touching exactly comes back UNDECIDED at that width, never SEPARATED, and the battery plants one.

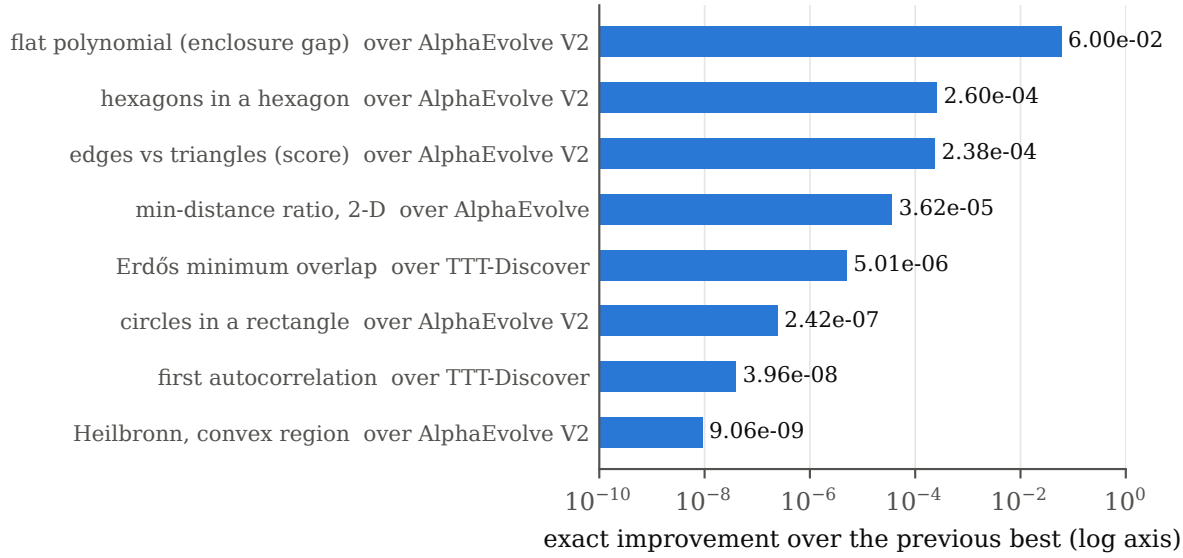


Figure 2: The 8 improvements over the previous best, each an exact difference of two rationals (for the flat polynomial, the gap between two disjoint certified enclosures), on a log axis. All are positive; the smallest, 9.06×10^{-9} on the Heilbronn configuration, is nine digits in and real. The largest exact difference, 2.6×10^{-4}, is the hexagon side; the benchmark score (edges vs triangles) is 2.38×10^{-4}.

Edges vs triangles is a benchmark score rather than a theorem: rows of twenty weights become (edge density, triangle density) points by the Newton identities, and the score is the area under the platform's slope-3 envelope plus ten times the largest gap in edge density. Recomputed exactly, both printed scores are the exact values' rounding (-0.712256 rounds -0.712256332697931), and the improvement is 2.38×10^{-4}. The Heilbronn and minimum-distance rows are ratios of exact areas and squared distances; their improvements, 9.06×10^{-9} and 3.62×10^{-5}, are the two smallest in the table and both are real.

Not decided, by name. Table 8 lists what the ledger refuses. A score estimated from ten million random samples is not a claim about a construction that exact arithmetic can decide, and rows of the README with no solution file at the pinned commit are NEEDS DATA: the day the bytes are published, they are rows. The ledger's battery runs 45 checks with 14 red controls (an overlapping pair, a box over the perimeter, a collinear triple, a repeated point, a value above 1, a zero row, a negative value, a planted touching pair), all firing.

problem	direction	previous best	exact difference	note
circles in a rectangle	maximise	AlphaEvolve V2	2.42×10^{-7}	ours taken as its REPAIRED exact witness
Heilbronn, convex region	maximise	AlphaEvolve V2	9.06×10^{-9}	
min-distance ratio, 2-D	minimise	AlphaEvolve	3.62×10^{-5}	
Erdős minimum overlap	minimise	TTT-Discover	5.01×10^{-6}	
edges vs triangles	maximise	AlphaEvolve V2	2.38×10^{-4}	
first autocorrelation	minimise	TTT-Discover	3.96×10^{-8}	
flat polynomial	minimise	AlphaEvolve V2	5.99×10^{-2}	a lower bound on the difference (rounded down): the two certified enclosures are disjoint
hexagons in a hexagon	minimise	AlphaEvolve V2	2.60×10^{-4}	both outer sides are exact decimals; both packings certified as witnesses

Table 7: Every improvement in the table, decided as an exact sign.

problem	the claim	status	why
prime-number-theorem	S(f) = 0.994179 (AlphaEvolve: 0.921292)	NOT DECIDABLE AS STATED	the platform's score is estimated from 10,000,000 random samples; a Monte Carlo score is not a mathematical claim about the construction and this ledger does not certify it.
tammes / second & third autocorrelation / uncertainty (README rows)	four further table rows	NEEDS DATA	the repository README lists them; no solution files are published in the repository at the pinned commit.

Table 8: The 2 entries of the table not decided here.

8 What is not claimed

No bound is improved anywhere in this paper and no construction was searched for except the free walk of Section 6, which beat nothing. Proposition 2 is not a correction of [7], which states the ring; it is the ring dependence decided from the artifact. Proposition 3 checks 18 of 496 nodes and says so; the remaining 478 rest on the author's reasoning, which we have no grounds to doubt and have not confirmed. Proposition 4 verifies the certificate the authors published, not the search that produced it, not their local optimality searches, and nothing about the minimum additive cost over all rank-23 schemes; 55 is an upper bound on a quantity whose true minimum nobody knows. The $\mathbb{F}_2$ statements of Section 6 need not lift to $\mathbb{Z}$ or to any other field, and no lifting was attempted. A REPAIRED row of Section 7 is not a refutation: a verifier's tolerance is a design choice (the platform has since dropped its circles slack, and its overlap verifier rescales rather than tolerates), and the repaired circles witness attains the printed value to nine of ten digits; the flat-polynomial enclosures certify the suprema of two published polynomials, not the flatness constant; and the benchmark score is defined by the platform's verifier rather than by mathematics. The sources audited are published, not peer-reviewed; the repository of Section 7 carries no licence file and its bytes are held here for verification only. Nothing here has been sent to any producer. Finally, the related rank-48 $\langle 4, 4, 4 \rangle$ scheme with non-complex coefficients derived from AlphaEvolve's (arXiv:2506.13242, as recorded in `notes/alphaevolve-48.md`) is a

different artifact and is not in the corpus.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records: `certs/strassen-certificate.json` (exported by `tools/export-strassen-certificate.js` at git `be82e12`), `corpus/strassen-corpus.json` and `corpus/alphaevolve-corpus.json` (the converted factors), `corpus/sources/PINS.json` with the pinned `npz` files, the notebook, the 55-addition certificate `add55-certificate.json` and Wang's two certificates `wang_cert_matrix_q02_n222.pb.txt`, `n333.pb.txt`; `certs/easota-ledger.json` with `corpus/easota/` (36 pinned files, digests in `meta.json`); `certs/bilinear-certificate.json` with `corpus/bilinear-bounds.json`. The instruments: `instruments/strassen` (`tensor.js`; family `families/strassen-audit.js`), `instruments/tensorlb` (`verify.py`), `instruments/slp`, `instruments/bilinear` with `machine/generate/` (targets, the flip walk), `instruments/easota` (`lib.js`, `decide.js`, `hexagons.js`). The commands: `node tools/paper-numbers/matmul-records.js` writes every macro of this document from those records and refuses if a record no longer supports a sentence — in the same run it re-certifies the fast-matmul corpus in-process, runs `python3 tools/verify_strassen.py certs/strassen-certificate.json` (no shared code), re-audits the addition certificate, re-runs both lower-bound audits, re-decides the 9 polynomial schemes, and runs the five batteries (`node instruments/strassen/battery.js`, `node instruments/slp/battery.js`, `python3 instruments/tensorlb/battery.py`, `node instruments/easota/battery.js`, `node instruments/bilinear/battery.js`), all in about ten seconds; `node tools/run-easota-ledger.js` rebuilds the EinsteinArena ledger from the pinned bytes; `node tools/run-bilinear-front.js --targets T8,C8` re-runs a walk; `instruments/hseva/.venv/bin/python tools/paper-numbers/matmul-records-figs.py` draws the two figures from the records; `node tools/build-paper-tex.js matmul-records` compiles. The report pages `reports/alphaevolve.html`, `reports/tensor-rank-bounds.html`, `reports/matmul-additions.html`, `reports/polynomial-multiplication.html` and `reports/easota.html` under `https://carlostoledo.co` are built from the same records by builders that re-run these gates and refuse on any deviation.

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